Engineering Materials

Chapter 1: Introduction



Objectives

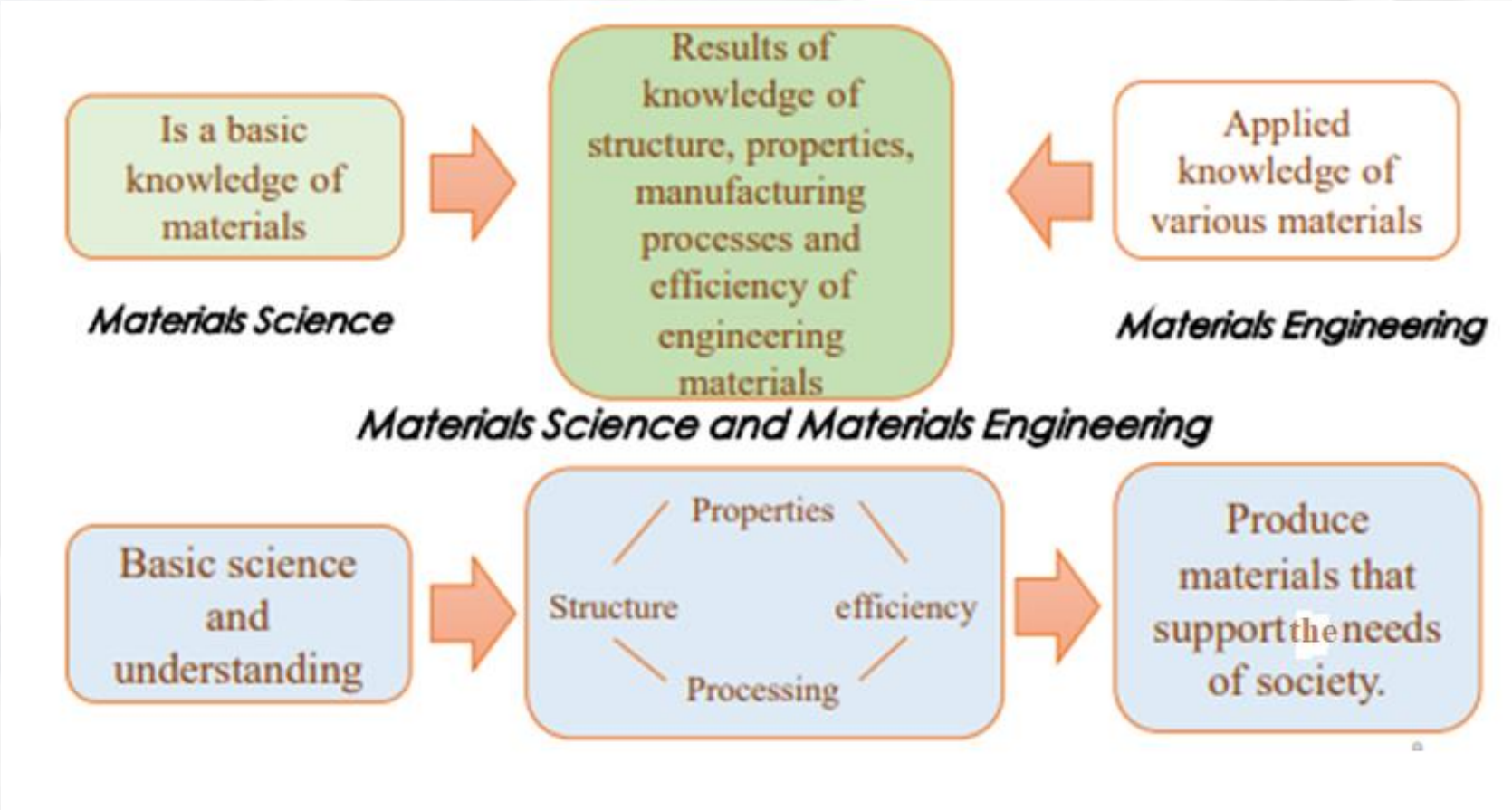
- Describe the concept of material science from historical viewpoint.
- State the importance of studying engineering materials.
- Identify the four main components of material science and their interrelation.
- Explain the various material classifications with examples.

Historical Background of Material Science

- The history of human civilization evolved from the Stone Age to the Bronze Age, Iron Age, Steel Age, and to the Space Age (synchronous with the Electronic/computer Age).
- Each age is marked by the advent of certain materials.
- The Iron Age brought tools and utensils
- The Steel Age brought railroads, instruments, and the Industrial Revolution.
- The Space/modern Age brought about the materials for stronger and light structures (e.g., composite materials)
- The Electronic Age brought semiconductors, and thus many varieties of electronic gadgets.

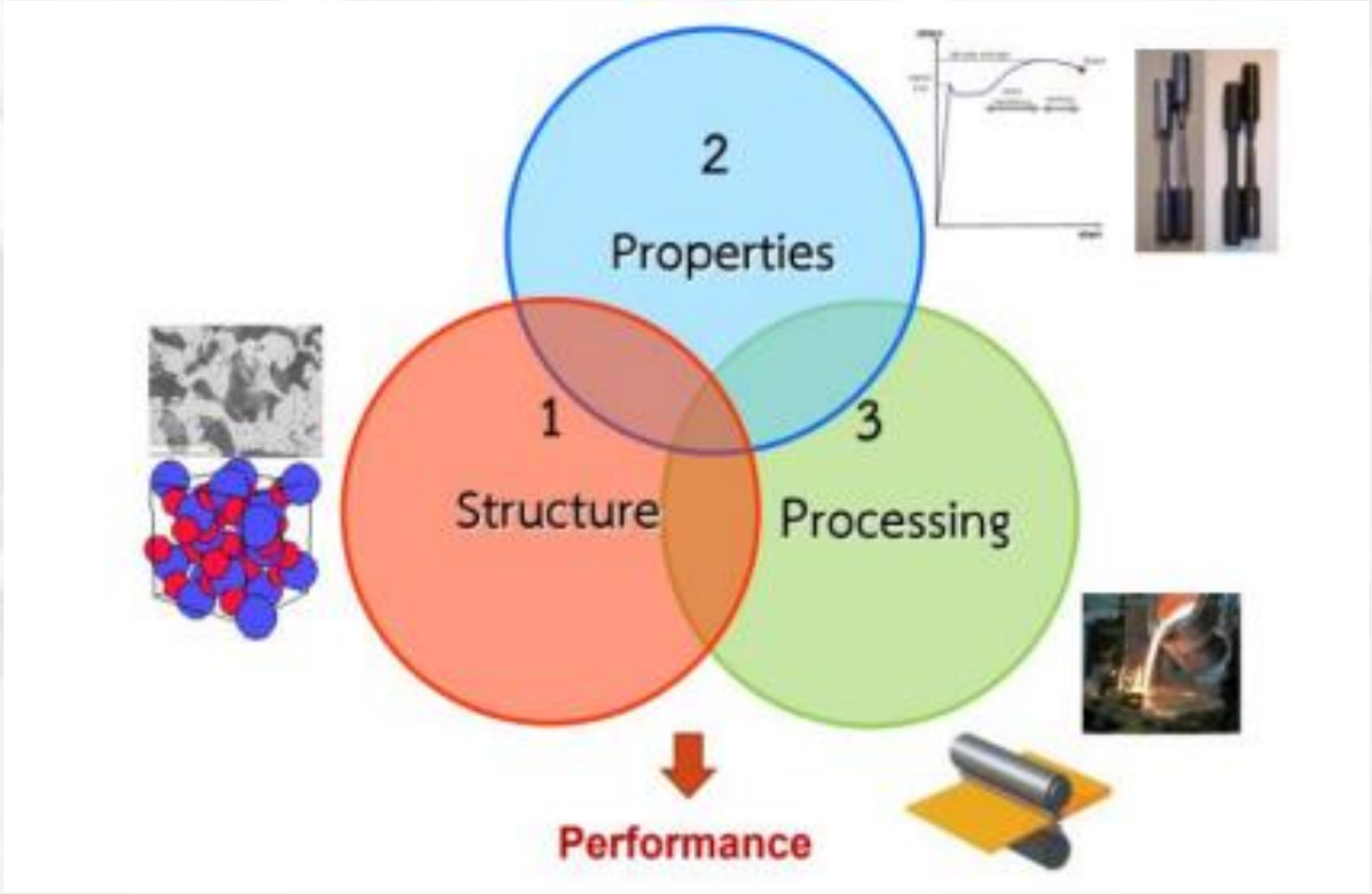


Materials Science and Engineering



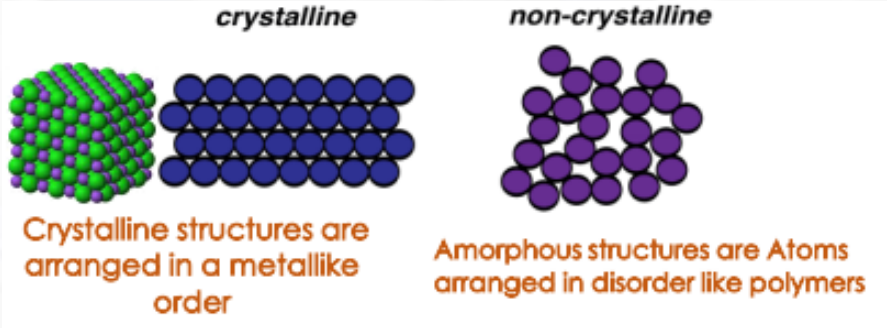
Materials Science and Engineering

- The behaviour of materials depend on three factors

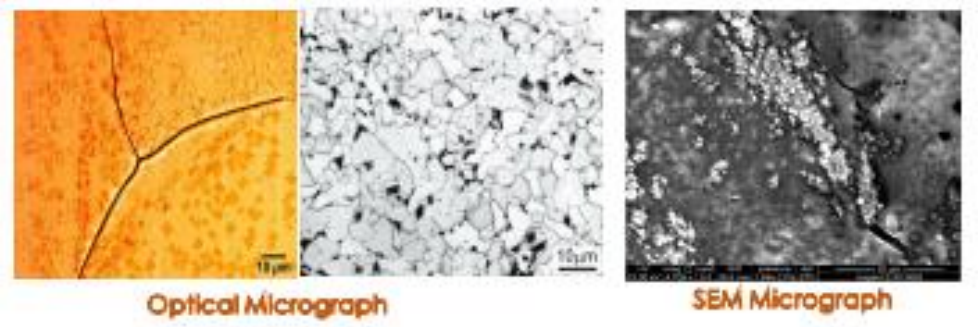


Structure of a material

- Atomic structure: Nucleus (Proton and Neutron) surrounded by electronic orbitals
- Crystal structure: Array characterization of atoms or molecules



- Microscopic structure: Can be viewed through a microscope.



- Macroscopic structure: Can be seen with the eyes.

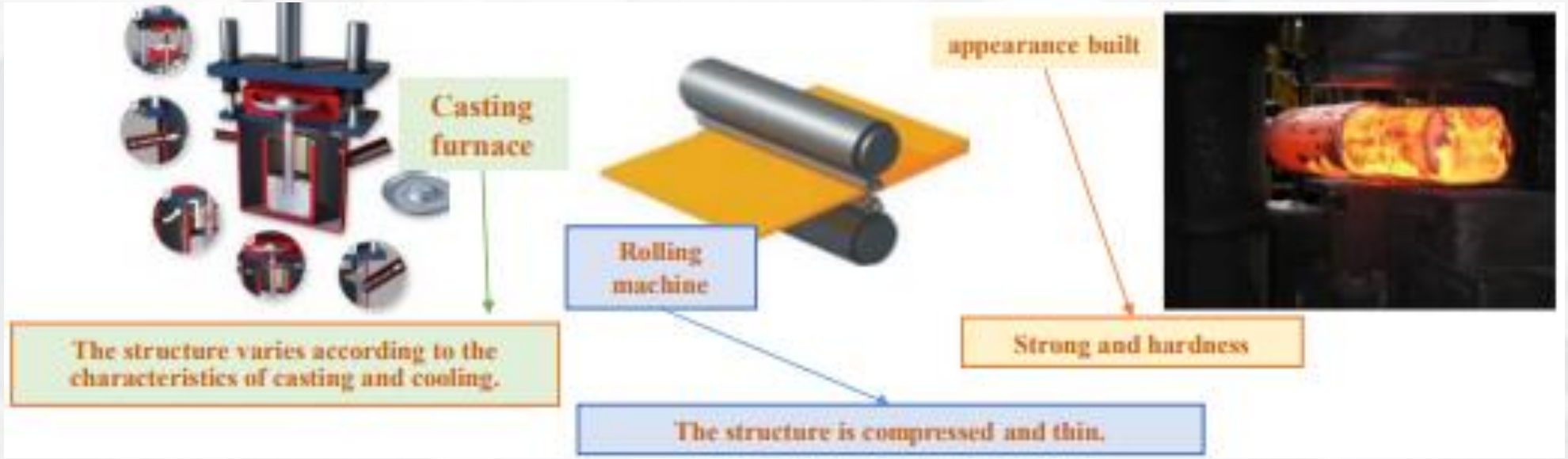
Properties

- The material's properties respond to the environment. The main engineering properties are:
- Chemical properties, such as structural features and composition of the compounds that make up the material
- Physical properties such as material adhesion, density, melting etc.
- Mechanical properties such as shrinkage, stiffness, hardness etc.
- Thermal properties such as heat conduction efficiency etc.
- Electrical properties such as electrical conductors etc.
- Magnetic properties such as magnetic field of the material.
- Optical properties such as the scattering of light, transparency of a material etc.



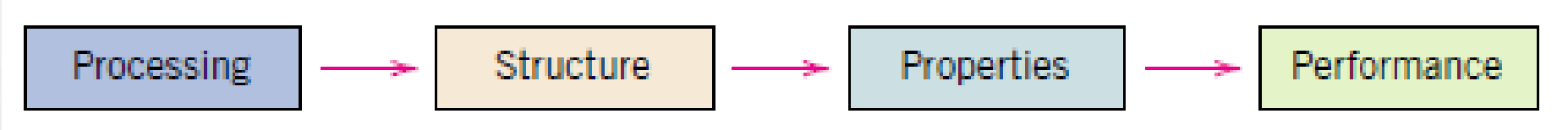
Processing

- Materials are processed by either heat or mechanical force.
- The result can be microstructural changes that can directly affect the material properties of a material.



Materials Science and Engineering

- Materials science has four (4) main components namely processing, structure, properties and performance.
- ***The structure*** of a material relates to the arrangement of its internal components. On an atomic level, structure encompasses the organization of atoms or molecules relative to one another.
- ***The property*** refers to the material kind and magnitude of response to a specific imposed stimulus. In general, definitions of properties are made independent of the material shape and size.
- The structure of a material will depend on how it is ***processed***.
- A material's ***performance*** will be a function of its properties.



Materials Science and Engineering

- Three thin disk specimens of aluminium oxide that have been placed over a printed page in order to demonstrate their differences in light-transmittance characteristics.
- The disk on the left is transparent whereas the one in the centre is translucent. The disk on the right is opaque.
- These differences in optical ***properties*** are a consequence of differences in ***structure*** of these materials, which have resulted from the way the materials were ***processed.***



Why Study Material Science and Engineering?

- Many engineers will at one time or another be exposed to a design problem involving materials.
- Many times, a materials problem is one of selecting the right material from the thousands that are available.
- The final decision is normally based on several criteria including: ***in-service conditions, material properties and cost.***
- The study of Materials Science and Engineering will help engineers to:

❖select a material for a given use based on considerations of cost and performance.

❖understand the limits of materials and the change of their properties with use.

❖be able to create a new material that will have some desirable properties.

❖be able to use the material for different application.



Classification of Materials

- Engineering materials can be grouped into
  - ❖Metals
  - ❖Ceramics
  - ❖Polymers
  - ❖Composites
  - ❖Semiconductors and
  - ❖Biomaterials



Metals

- Metals are characterized by high thermal and electrical conductivity.
- Metals are strong yet deformable under applied mechanical loads; opaque to light (shiny if polished).
- Atoms are bound together by metallic bonds and weaker Van der Waals forces
- Pure metals are not good enough for many applications, especially structural applications.
- Metals are used in alloy form.



Fig. Objects made of metals and metal alloys

Ceramics

- These are inorganic compounds, and usually made either of oxides, carbides, nitrides, or silicates of metals.
- Ceramics are typically partly crystalline and partly amorphous.
- Atoms (ions often) in ceramic materials behave mostly like either positive or negative ions, and are bound by very strong Coulomb forces between them.
- They are characterized by very high strength under compression, low ductility; usually insulators to heat and electricity. Examples: glass, porcelain, many minerals.



Fig. Objects made of ceramics

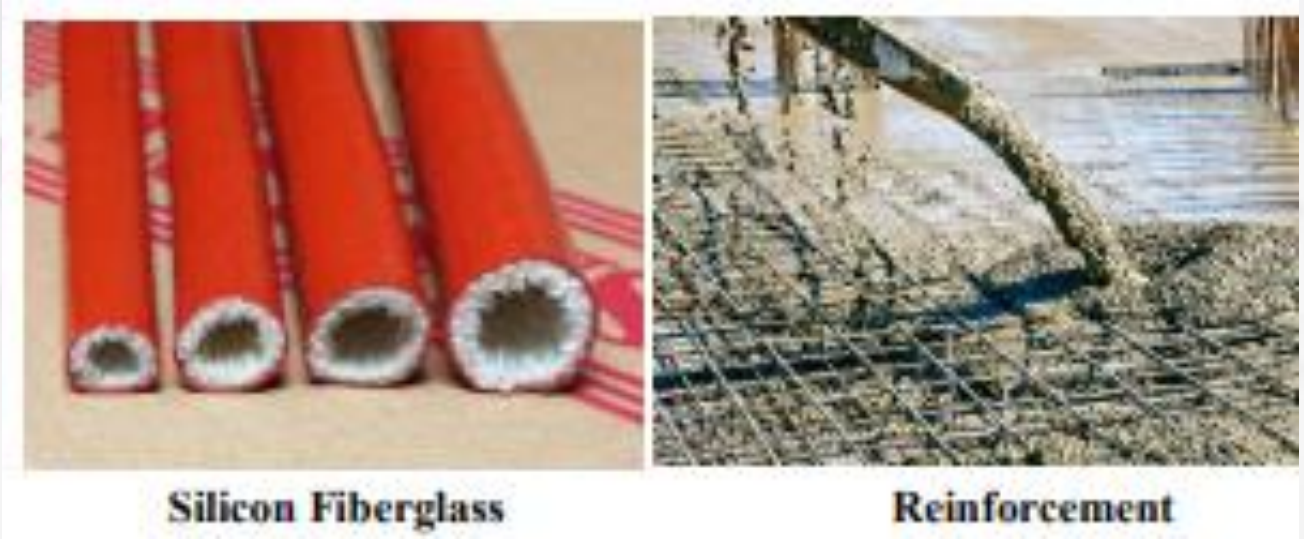
Polymers

- Polymers in the form of thermo-plastics (nylon, polyethylene, polyvinyl chloride, rubber, etc.) consist of molecules that have covalent bonding within each molecule and van der Waals forces between them.
- Polymers in the form of thermo-sets (e.g., epoxy, phenolics, etc.) consist of a network of covalent bonds.
- Polymers are amorphous, except for a minority of thermoplastics.
- Due to the kind of bonding, polymers are typically electrical and thermal insulators.
- Conducting polymers can be obtained by doping, and conducting polymer-matrix composites can be obtained by the use of conducting fillers.
- They decompose at moderate temperatures of about 100 to 400 °C, and are lightweight.



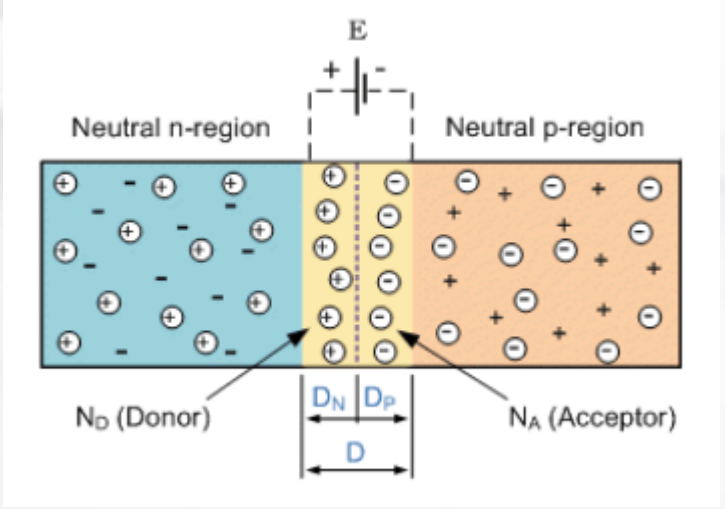
Composite Materials

- Composed of two (or more) individual materials, which come from the categories previously discussed - metals, ceramics, and polymers.
- The design goal of a composite is to achieve a combination of properties that is not displayed by any single material, and also to incorporate the best characteristics of each of the component materials.
- There are some natural composites available in nature, for example wood.



Advanced Materials: Semiconductors

- Semiconductors have electrical properties that are intermediate between the electrical conductors (i.e., metals and metal alloys) and insulators (i.e., ceramics and polymers).
- Properties: when the temperature rises, the conductivity of the semiconductor Will be better (as opposed to metal)
- Example: Silicon chip and micro-electronic devices (microelectronic devices), which is important in the production of satellites, computer, calculator, digital watch, robots etc.
- Semiconductors have made possible the advent of integrated circuitry that has totally revolutionized the electronics and computer industries over the past three decades.





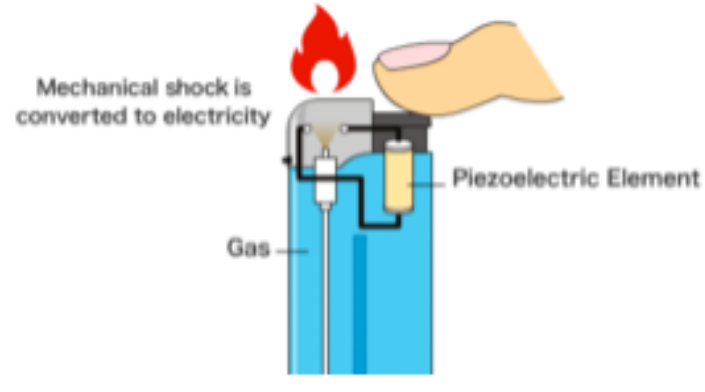
Advanced materials: Biomaterials

- These are materials used for replacement of damaged or diseased human body parts.
- They must be biocompatible with body tissues, and must not produce toxic substances.
- Other important material factors are: ability to support forces; low friction, wear, density, cost; reproducibility.
- Typical applications involve heart valves, hip joints, dental implants, intraocular lenses.
- Examples: Stainless steel, Co-28Cr-6Mo, Ti-6Al-4V, ultra-high molecular weight poly-ethylene, high purity dense Al-oxide, etc.



Smart Materials

- *Smart* (or *intelligent*) *materials* are a group of new and state-of-the-art materials now being developed that will have a significant influence on many of our technologies.
- The adjective *smart* implies that these materials are able to sense changes in their environment and then respond to these changes in predetermined manners.
- In addition, this “smart” concept is being extended to rather sophisticated systems that consist of both smart and traditional materials.

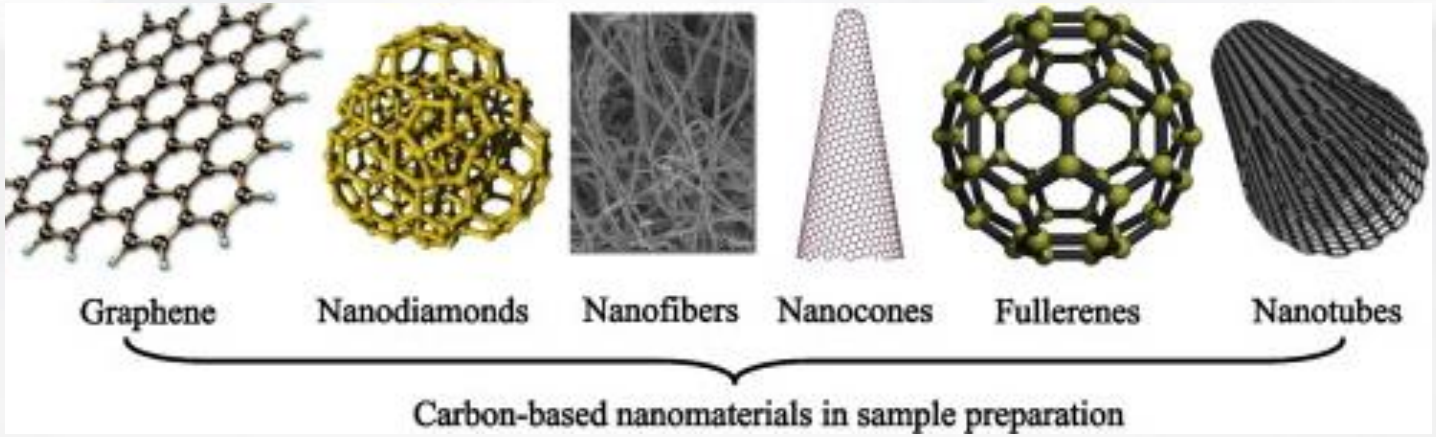



Shape Memory Alloy Shirt and Jacket (2010)


Smart fabrics and power laces

Nanomaterials

- Nanomaterials have fascinating properties and tremendous technological promise.
- They may be any one of the four basic types - metals, ceramics, polymers, and composites.
- They are classified on the basis of their size.
- The *nano* -prefix denotes that the dimensions of these structural entities are on the order of a nanometer (10^{-9}m)
- Some of the physical and chemical characteristics exhibited by matter may experience dramatic changes as particle size approaches atomic dimensions.
- Nanomaterials are finding niches in electronic, biomedical, sporting, energy production, and other industrial applications



Application of Engineering Materials



1. Car industry



2. Computer electronics industry



3. Construction industry



4. Water industry



5.Aircraft industry

Modern Materials' needs

- Some typical needs of modern materials include:
  - ❖Engine efficiency increases at high temperatures: requires high temperature structural materials
  - ❖Use of nuclear energy requires solving problem with residues, or advances in nuclear waste processing.
  - ❖Hypersonic flight requires materials that are light, strong and resist high temperatures.
  - ❖Optical communications require optical fibres that absorb light negligibly.
  - ❖Civil construction – materials for unbreakable windows.
  - ❖Structures: materials that are strong like metals and resist corrosion like plastics.

Assignment

- Identify five (3) smart materials from open literature, and explain how they can be applied in your field of study.